

Bounds for hypergraph Universality at D=1

Sifei Li

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For any log without indicating its base, we assume it is base on 2.

1 Prerequisites

Theorem 1 (Chernoff bound). *Let $X = \sum_{i=1}^n X_i$, where the X_i are independent Bernoulli random variables. Then*

1. $\mathbb{P}(X \geq (1 + \delta)\mathbb{E}X) \leq \exp\left(-\frac{\delta^2}{2+\delta}\mathbb{E}X\right)$ for all $\delta > 0$,
2. $\mathbb{P}(X \leq (1 - \delta)\mathbb{E}X) \leq \exp\left(-\frac{\delta^2}{2}\mathbb{E}X\right)$ for all $0 < \delta < 1$.

The observation 2.1 in the paper [Kat25] still hold for $D = 1$, which is a 1-degenerate r -graph has at most rn/k vertices of degree at least k .

2 Proofs

Definition 1 (The block model for the case $D = 1$). *Fix $r \geq 3$, and let $q := r - 1$. Let N be the smallest integer such that*

$$n^{q^{1-N}} \leq 3^{q^2}.$$

For $1 \leq k \leq N$, define

$$\Delta_k := n^{q^{-k}},$$

and set $\Delta_0 := n$, so $\Delta_{k-1} = \Delta_k^q$. Let $W = W_1 \sqcup \cdots \sqcup W_N$ be a disjoint union of blocks satisfying

$$|W_k| = \frac{100 \cdot 3^{qr}n}{\Delta_k} \quad \text{for each } 1 \leq k \leq N.$$

Each block W_k is further partitioned into sub-blocks

$$W_k = W_{k,1} \sqcup \cdots \sqcup W_{k,\log n},$$

where

$$|W_{k,1}| = \frac{1}{2}|W_k| \quad \text{and} \quad |W_{k,j}| \geq \frac{|W_k|}{2 \log n} \quad \text{for all } j \geq 2.$$

For $s \in \binom{W}{r}$, let $\pi_i(s) := |s \cap W_i|$, and define

$$p^* := K_r \frac{(\log n)^2}{n}, \quad p_s := \min \left\{ 1, p^* \prod_{i=1}^N \Delta_i^{\pi_i(s)} \right\},$$

where $K_r = \max\{1, \frac{8}{25} \cdot \ln 2 \cdot 3^{-q}(1 + \frac{4}{r})\}$ is a constant depending only on r . The formula of K_r comes from the proof of Lemma 2. The random r -graph Γ on vertex set W is obtained by including each $s \in \binom{W}{r}$ as its edge independently with probability p_s .

Proposition 1. Fix $r \geq 3$, for all sufficiently large n , all the definitions follow Definition 1. The followings hold: *(may insert more if needed)*

1. $N < \log \log n$

Proof. from the definition of N in Definition 1, we have $n^{q^{1-(N-1)}} > 3^{q^2}$. Taking $\log_3$ and $\log_n$ on both sides gives $(2 - N) + \log_q \log_3 n > 2$. Rearranging the equation gives:

$$N < \log_q \log_3 n \leq \log \log n$$

as desired. $\square$

Proposition 2 (The $D = 1$ version of Proposition 4.2(c)). For every fixed $r \geq 3$, there is a constant $C_r > 0$ such that, for all sufficiently large n , the random r -graph $\Gamma = \Gamma(r, n, 1)$ defined in Definition 1 has at most $150 \cdot 3^q r n = 150 \cdot 3^{r-1} r n$ vertices, and asymptotically almost surely has at most

$$C_r (\log \log n)^r (\log n)^2 n^{r-1}$$

edges with some constant C_r depending on r .

Proof. We first bound the number of vertices. By the definition of N , we have

$$\Delta_{N-1} = n^{q^{1-N}} \leq 3^{q^2} \text{ and } \Delta_{N-2} = n^{q^{2-N}} > 3^{q^2}.$$

Raise the left equation to the power $1/q$ and combine with the right equation,

$$3^q \geq \Delta_{N-2}^{1/q^2} = \Delta_N > 3.$$

Moreover, using the property $\Delta_{k-1} = \Delta_k^q$ for $0 \leq k \leq N$, and $q = r - 1 \geq 2$, we get

$$\begin{aligned} |W| &= |W_1| + |W_2| + \dots + |W_N| = 100 \cdot 3^q r n \sum_{k=1}^N \frac{1}{\Delta_k} = 100 \cdot 3^q r n \sum_{s=0}^{N-1} \frac{1}{\Delta_N^{q^s}} \\ &\leq 100 \cdot 3^q r n \sum_{s=0}^{\infty} \frac{1}{\Delta_N^{q^s}} \leq 100 \cdot 3^q r n \sum_{s=0}^{\infty} \frac{1}{3^{2^s}} = 100 \cdot 3^q r n \frac{1}{1 - \frac{1}{3}} = 150 \cdot 3^q r n. \end{aligned}$$

Thus Γ has at most $150 \cdot 3^q r n$ vertices.

We now estimate the expected number of edges. Fix an ordered r -tuple

$$S = (i_1, \dots, i_r) \in [N]^r.$$

For each $k \in [N]$, let $m_S(k)$ be the multiplicity of k in S . The expected number of edges of Γ with one vertex in each of $W_{i_1}, \dots, W_{i_r}$ is at most

$$\min \left\{ 1, p^* \prod_{\ell=1}^r \Delta_{i_\ell} \right\} \prod_{k \in [N]} |W_k|^{m_S(k)} \leq p^* \prod_{\ell=1}^r \Delta_{i_\ell} \prod_{k \in [N]} |W_k|^{m_S(k)}.$$

Thus

$$\mathbb{E}[|E(W_{i_1}, \dots, W_{i_r})|] \leq p^* \prod_{\ell=1}^r \Delta_{i_\ell} \prod_{k \in [N]} |W_k|^{m_S(k)} = p^* \prod_{\ell=1}^r \Delta_{i_\ell} \frac{(100 \cdot 3^q r n)^2}{\Delta_{i_\ell}} = p^* \prod_{\ell=1}^r 100 \cdot 3^q r n$$

Summing over all N^r choices of r -tuples and using $N \leq \log \log n$ from Proposition 1, we get

$$\mathbb{E}|E(\Gamma)| \leq N^r p^* \prod_{\ell=1}^r 100 \cdot 3^q r n = N^r p^* (100 \cdot 3^q r n)^r \leq (\log \log n)^r \frac{K_r (\log n)^2}{n} (100 \cdot 3^q r n)^r := C'_r (\log \log n)^r (\log n)^2 n^{r-1},$$

where $C'_r = K_r(100 \cdot 3^q \cdot r)^r$.

It remains to turn this expectation bound into an asymptotically almost sure bound. Denote $M = C'_r(\log \log n)^r(\log n)^{2n^{r-1}}$ and apply part 1 of Theorem 1 (Chernoff bound) with

$$\delta = 2 \frac{M}{\mathbb{E}|E(\Gamma)|} - 1,$$

notice that $\delta \geq 1$, we get

$$\begin{aligned} & \mathbb{P}[|E(\Gamma)| \geq 2M] \\ &= \mathbb{P}(|E(\Gamma)| \geq (1 + \delta)\mathbb{E}|E(\Gamma)|) \\ &\leq \exp\left[-\frac{\delta^2}{2 + \delta}\mathbb{E}|E(\Gamma)|\right] \\ &\leq \exp\left[-\frac{\delta\mathbb{E}|E(\Gamma)|}{3}\right] \\ &\leq \exp[-M/3] \\ &= o(1) \end{aligned}$$

Taking $C_r = 2C'_r = 2K_r(100 \cdot 3^q \cdot r)^r$ finishes the proof. $\square$

Definition 2 (Embedding Strategy). *[Kat25] Given any D -degenerate r -graph H , let $v_1, \dots, v_n$ be a D degeneracy ordering of $V(H)$, and let Γ have the properties of Proposition 2. Our embedding strategy is defined by the iterative construction of a partial embedding $\psi_t : \{v_1, \dots, v_t\} \rightarrow \Gamma$. We start with ψ_0 , the trivial partial embedding of no vertices into Γ .*

For each $1 \leq t \leq n$, we say a vertex $u \in \Gamma$ is a candidate for v_t if $\psi_{t-1}(L^-(v_t)) \subseteq L(u)$. We say such a vertex $u \in \Gamma$ is an available candidate if in addition $u \notin \text{Im}(\psi_{t-1})$. Let k be minimal such that $\Delta_k < \deg(v_t)$. If there is no such k choose $k = N$. Choose j minimal such that there is some $u \in W_{k,j}$ which is an available candidate for v_t . We define $\psi_t = \psi_{t-1} \cup \{v_t \rightarrow u\}$. If there is no available candidate for v_t in W_k , we say ψ_t and the subsequent partial embeddings do not exist and that the embedding strategy fails at time t . If the embedding strategy does not fail, then $\psi := \psi_n$ gives an embedding of H into Γ .

This embedding strategy will not change for the case $D = 1$, same as the definition of well-behaved multisets.

Definition 3 (Well-behaved multisets for the case $D = 1$). *Let $\mathcal{B} = (B_1, \dots, B_t)$ be a multiset such that each $B_i \subseteq \binom{W}{r-1}$. For $B_i \subseteq \binom{W}{r-1}$, write*

$$V(B_i) := \bigcup_{b \in B_i} b,$$

and write

$$V(\mathcal{B}) := \bigcup_{i=1}^t V(B_i).$$

We say that $\mathcal{B}$ is well-behaved if the following three conditions hold:

$$|B_i| \leq 1 \quad \text{for all } i \in [t], \tag{WB1}$$

$$|\{i \in [t] : u \in V(B_i)\}| \leq \Delta_{k-1} \quad \text{for all } 1 \leq k \leq N \text{ and all } u \in W_k, \tag{WB2}$$

and

$$|V(\mathcal{B}) \cap W_{k,j}| \leq \frac{1}{2}|W_{k,j}| \quad \text{for all } 1 \leq k \leq N \text{ and all } 1 \leq j \leq \log n. \tag{WB3}$$

Lemma 1. *Let $r \geq 3$, let n be sufficiently large, and let $t \in [n]$. Let Γ be the random block model defined in Definition 1, and let $\mathcal{B} = (B_1, \dots, B_t)$ be a well-behaved multiset of size t . Fix $1 \leq k \leq N$ and $v \in W_k \setminus V(\mathcal{B})$. Let $\mathcal{E}_v$ be the event that there exists some $B_i \in \mathcal{B}$ such that $B_i \subseteq L(v)$. Then*

$$\mathbb{P}(\mathcal{E}_v) \geq \min \left\{ \frac{1}{2}, \frac{1}{2} tp^* \Delta_k \right\}.$$

Proof. Let $1 \leq k \leq N$, and $v \in W_k \setminus V(\mathcal{B})$. If $B_i = \emptyset$ for some $i \in [t]$, then $B_i \subseteq L(v)$ always holds, and hence $\mathbb{P}(\mathcal{E}_v) = 1$. We may therefore assume that $B_i \neq \emptyset$ for every $i \in [t]$. By (WB1), for each $i \in [t]$ there is an $(r-1)$ -set $b_i \in \binom{W}{r-1}$ such that $B_i = \{b_i\}$.

Let $S := \{b_i : i \in [t]\}$ be the set of distinct $(r-1)$ -sets appearing among the b_i 's, and for $b \in S$, let $m_b := |\{i \in [t] : b_i = b\}|$ be the multiplicity of b , and we have $\sum_{b \in S} m_b = t$.

Fix $b \in S$, and write $b = \{x_1, \dots, x_{r-1}\}$, where $x_l \in W_{k_l}$ for $1 \leq l \leq r-1$. Condition (WB2) gives

$$m_b \leq |\{i \in [t] : x_l \in V(B_i)\}| \leq \Delta_{k_{\ell}-1} \quad \text{for each } 1 \leq \ell \leq r-1.$$

For each $1 \leq k_l \leq N$, $\Delta_{k_l} = \Delta_{k_l-1}^{1/q} \geq m_b^{1/q}$. Therefore

$$\prod_{l=1}^{r-1} \Delta_{k_l} \geq m_b^{(r-1)/q} = m_b.$$

Let E_b be the event that $b \in L(v)$. Since $v \notin V(\mathcal{B})$, the set $\{v\} \cup b$ has size r , and E_b is exactly the event that $\{v\} \cup b$ is an edge of Γ . Therefore

$$\mathbb{P}(E_b) = p_{\{v\} \cup b} = \min \left\{ 1, p^* \Delta_k \prod_{\ell=1}^q \Delta_{k_{\ell}} \right\} \geq \min\{1, p^* \Delta_k m_b\}.$$

The events $(E_b)_{b \in S}$ are independent, since they concern distinct r -sets of the form $\{v\} \cup b$. Thus, $\mathcal{E}_v = \bigcup_{b \in S} E_b$, and $\mathbb{P}(\mathcal{E}_v) = 1 - \prod_{b \in S} (1 - \mathbb{P}(E_b))$.

Furthermore, notice that

$$\sum_{b \in S} \mathbb{P}(E_b) \geq \sum_{b \in S} \min\{1, p^* \Delta_k m_b\} \geq \min \left\{ 1, p^* \Delta_k \sum_{b \in S} m_b \right\} = \min\{1, tp^* \Delta_k\}.$$

Since $1 - x \leq e^{-x}$ for all $x \in [0, 1]$ (which follows from the Taylor expansion $e^{-x} = \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} = 1 - x + \frac{x^2}{2} - \dots \geq 1 - x$), we have

$$\mathbb{P}(\mathcal{E}_v) \geq 1 - \exp \left(- \sum_{b \in S} \mathbb{P}(E_b) \right) \geq 1 - \exp(-\min\{1, tp^* \Delta_k\}).$$

Finally, notice that for every $x \geq 0$, $1 - e^{-\min\{1, x\}} \geq \frac{1}{2} \min\{1, x\}$, which can be verified by considering two cases: If $x \leq 1$, then $1 - e^{-x} \geq x - \frac{x^2}{2} \geq \frac{x}{2}$ by Taylor expansion again; if $x \geq 1$, then $1 - e^{-1} > \frac{1}{2}$. Applying this with $x = tp^* \Delta_k$, we get

$$\mathbb{P}(\mathcal{E}_v) \geq \frac{1}{2} \min\{1, tp^* \Delta_k\} = \min \left\{ \frac{1}{2}, \frac{1}{2} tp^* \Delta_k \right\},$$

as required. $\square$

Lemma 2. *Let $r \geq 3$ and n be sufficiently large. Then Γ defined in the Definition 1 a.a.s. satisfies the following: $\forall t \leq n$ and well-behaved multiset $\mathcal{B}$ of size t . $\forall k \leq N, j \leq \log n$, we have*

$$|\{u \in W_{k,j} : \exists B_i \in \mathcal{B}, B_i \subseteq L(u)\}| \geq \min \left\{ \frac{1}{16}, \frac{tp^* \Delta_k}{8} \right\} |W_{k,j}| \quad (1)$$

Proof. Let r, n, k, j be integers. $r \geq 3$, n large enough. Let $t \leq n$ and $\mathcal{B}$ be a well-behaved multiset of size t . Fix $1 \leq k \leq N$ and $1 \leq j \leq \log n$. If $tp^*\Delta_k > 1$, then we can pick a t' such that $\frac{1}{2} < t'p^*\Delta_k \leq 1$, and arbitrarily choose $\mathcal{B}' \subseteq \mathcal{B}$ with $|\mathcal{B}'| = t'$. Then

$$|\{u \in W_{k,j} : \exists B_i \in \mathcal{B}, B_i \subseteq L(u)\}| \geq \frac{1}{16}|W_{k,j}|$$

as desired. Notice that such t' exists as $p^*\Delta_k$ is small enough, indeed, $p^*\Delta_k \leq K_r(\log n)^2 n^{(1/q)-1} = o(1)$ with $q \geq 2$. Thus we first prove the lemma under the assumption $tp^*\Delta_k \leq 1$, then the large case follows by applying the already-proved small case to the sub-multiset $\mathcal{B}'$.

If there exists an i such that $B_i = \emptyset$, then $B_i \subseteq L(u)$ for every $u \in W_{k,j}$, and the desired inequality is immediate.

Let $U := W_{k,j} \setminus V(\mathcal{B})$. By (WB3), $|U| \geq \frac{1}{2}|W_{k,j}|$. For each $u \in U$, let X_u be the indicator of the event $\mathcal{E}_u$ that there exists $i \in [t]$ such that $B_i \subseteq L(u)$, and set $X := \sum_{u \in U} X_u$.

The random variables $(X_u)_{u \in U}$ are independent. Indeed, X_u depends only on edge indicators of sets of the form $\{u\} \cup b$, where b is the unique element of some B_i . By definition, the edge indicator is an independent random variable, so X_u is also an independent random variable.

By Lemma 1, for every $u \in U$, $\mathbb{P}(\mathcal{E}_u) \geq \min\{\frac{1}{2}, \frac{1}{2}tp^*\Delta_k\}$. Since $tp^*\Delta_k \leq 1$, this gives $\mathbb{P}(\mathcal{E}_u) \geq tp^*\Delta_k/2$. Therefore,

$$\mathbb{E}X = \sum_{u \in U} \mathbb{E}X_u \geq \frac{1}{2}tp^*\Delta_k|U| \geq \frac{1}{4}tp^*\Delta_k|W_{k,j}| := 2T. \quad (2)$$

where $T = \frac{1}{8}tp^*\Delta_k|W_{k,j}|$.

Now we show that the probability that the conclusion fails is $o(1)$. By part 2 of Theorem 1, $\Pr(X \leq T) \leq \Pr(X \leq (1 - \frac{1}{2})\mathbb{E}X) \leq \exp(-\frac{1}{8}\mathbb{E}X)$. To obtain a lower bound on $\mathbb{E}X$, note that $|W_{k,j}| \geq \frac{|W_k|}{2 \log n} = \frac{50 \cdot 3^q r n}{\Delta_k \log n}$, therefore

$$\mathbb{E}(X) \geq \frac{1}{4}tp^*\Delta_k|W_{k,j}| \geq \frac{1}{4}tp^*\Delta_k \frac{50 \cdot 3^q r n}{\Delta_k \log n} \geq 25/2 \cdot 3^q r K_r t \log n.$$

Combining all together and using $\log n = \frac{\ln n}{\ln 2}$,

$$\Pr(X < T) \leq \exp\left(-\frac{1}{8} \cdot 25/2 \cdot 3^q r K_r t \log n\right) = n^{-25 \cdot 3^q \cdot r \cdot K_r \cdot t / (16 \ln 2)} = n^{-\alpha t}$$

where $\alpha = 25 \cdot 3^q \cdot K_r / (16 \ln 2)$. Finally, we count the number of choices. By Proposition 2, $|V(\Gamma)| \leq 150 \cdot 3^q r n$. Thus each B_i has at most $\binom{|V(\Gamma)|}{r-1} + 1 \leq (150 \cdot 3^q r)^r n^r$ possible choices.

Therefore the number of well-behaved multisets $\mathcal{B}$ of size t is at most $(150 \cdot 3^q r)^{rt} n^{rt}$. By Proposition 1, there are at most $N \leq \log \log n$ choices for k , $\log n$ choices for j .

Hence the probability that Lemma 2 fails for some fixed t is at most

$$(\log \log n)(\log n) (150 \cdot 3^q r)^{rt} n^{rt} n^{-\alpha t}.$$

Recall that we picked K_r sufficiently large so that

$$r(1 - \alpha) \leq -5,$$

then

$$(\log \log n)(\log n) (150 \cdot 3^q r)^{rt} n^{rt} n^{-\alpha t} \leq (150 \cdot 3^q r)^{rt} (\log \log n)(\log n) n^{-5t} \leq (\log \log n)(\log n) n^{-4t}$$

using the fact that $(150 \cdot 3^q r)^r \leq n$, as n is sufficiently large.

Finally, taking a union bound over all $t \leq n$,

$$\Pr(\text{Lemma 2 fails}) \leq \sum_{t=1}^n (\log \log n)(\log n) n^{-4t} \leq (\log \log n)(\log n) n^{-4} \cdot n \leq n^{-1} = o(1).$$

This completes the proof. $\square$

Theorem 2. For $r \geq 3$, there exists a constant C_r such that for all sufficiently large n , there exists an r -uniform hypergraph Γ with

$$e(\Gamma) \leq C_r n^{r-1} (\log n)^2 (\log \log n)^r$$

which is universal for the class $\mathcal{H}^{(r)}(n, 1)$ of n -vertex 1-degenerate r -graphs.

Proof. Let $r \geq 3$. By Proposition 2, the random r -graph Γ defined in Definition 1 asymptotically almost surely satisfies

$$e(\Gamma) \leq C_r n^{r-1} (\log n)^2 (\log \log n)^r,$$

and the statement of Lemma 2. It therefore remains to show that Γ is $\mathcal{H}^{(r)}(n, 1)$ -universal.

Let $H \in \mathcal{H}^{(r)}(n, 1)$. Since H is 1-degenerate, there exists an ordering $v_1, \dots, v_n$ of $V(H)$ such that every vertex v_t has at most one back-link. We embed H into Γ using the embedding strategy from Definition 2.

Suppose, for a contradiction, that Γ is not $\mathcal{H}^{(r)}(n, 1)$ -universal. Then the embedding strategy must fail at some time t^* for some H . This means that there is no available candidate for v_{t^*} .

Let k^* be the positive integer chosen by the embedding strategy. We have $\Delta_{k^*} \leq d_H(v_{t^*}) \leq \Delta_{k^*-1}$ if $k^* < N$ and $d_H(v_{t^*}) \leq \Delta_{N-1}$ if $k^* = N$, where $d_H(v_{t^*})$ is the degree of v_{t^*} in H .

Let $B := \{\psi_{t^*-1}(L^-(v_{t^*}))\}$, notice that B is empty or contains a single $(r-1)$ -set. We claim that $\{B\}$ is a well-behaved multiset. If $B = \emptyset$, then $\{B\}$ is a well-behaved multiset straightforwardly. If $|B| = 1$, condition (WB1) is immediate. For (WB2), let $1 \leq k \leq N$ and let $u \in W_k$. Since $|B| = 1$, $|\{i \in [1] : u \in V(B_i)\}| \leq 1 < 3 \leq \Delta_N \leq \Delta_{k-1}$, thus (WB2) holds.

For (WB3), let $1 \leq k \leq N$ and $1 \leq j \leq \log n$. Since B consists of a single $(r-1)$ -set, $|V(B) \cap W_{k,j}| \leq r-1$. On the other hand,

$$|W_{k,j}| \geq \frac{|W_k|}{2 \log n} = \frac{50 \cdot 3^q r n}{\Delta_k \log n} = \frac{50 \cdot 3^q r n^{1-q^{-k}}}{\log n}.$$

Since $q = r-1 \geq 2$, we have $1 - q^{-k} > 0$. For all sufficiently large n ,

$$|V(B) \cap W_{k,j}| \leq \frac{1}{2} \frac{50 \cdot 3^q r n^{1-q^{-k}}}{\log n} = \frac{1}{2} |W_{k,j}|$$

. Hence (WB3) also holds. Therefore $\{B\}$ is well-behaved.

Applying Lemma 2 with the well-behaved multiset $\{B\}$, $k = k^*$, and $j = \log n$, we obtain

$$\begin{aligned} |\{u \in W_{k^*, \log n} : B \subseteq L(u)\}| &\geq \min \left\{ \frac{1}{16}, \frac{1}{8} p^* \Delta_{k^*} \right\} |W_{k^*, \log n}| \\ &= \min \left\{ \frac{1}{16} \cdot \frac{100 \cdot 3^q r n}{2 \Delta_{k^*} \log n}, \frac{1}{8} \cdot \frac{100 \cdot 3^q r n}{2 \Delta_{k^*} \log n} \cdot \frac{K_r (\log n)^2}{n} \Delta_{k^*} \right\} \\ &= \min \left\{ \frac{100 \cdot 3^q r}{32} \cdot \frac{n^{1-q^{-k^*}}}{\log n}, \frac{50 \cdot 3^q r K_r}{8} \log n \right\} \\ &> 1 \end{aligned}$$

The last inequality uses the facts that $1 - q^{-k^*} > 0$ and both quantities tend to infinity as n being sufficiently large. Thus there exists more than one candidate for v_{t^*} in $W_{k^*, \log n}$. Since the embedding strategy fails at time t^* , every such candidate must belong to $\text{Im}(\psi_{t^*-1})$.

Define

$$L_{k,j} := (4 \log n)^{1-j} \begin{cases} \frac{r n}{\Delta_k}, & 1 \leq k \leq N-1, \\ n, & k = N. \end{cases}$$

We claim that for every $t \leq t^* - 1$, $|\text{Im}(\psi_t) \cap W_{k,j}| \leq L_{k,j}$ for all $1 \leq k \leq N$ and $1 \leq j \leq \log n$.

Assuming this claim is true, we obtain

$$\begin{aligned}
|\text{Im}(\psi_{t^*-1}) \cap W_{k^*, \log n}| &\leq L_{k^*, \log n} \\
&= (4 \log n)^{1-\log n} \frac{rn}{\Delta_{k^*}} \\
&= (4 \log n)^{1-\log n} r n^{1-q^{-k^*}} \\
&= 4r \log n \cdot n^{1-q^{-k^*}-\log(4 \log n)} \\
&= 4r \log n \cdot n^{1-q^{-k^*}-2-\log \log n} \\
&\leq 4r \log n \cdot n^{-1} \\
&\leq 1,
\end{aligned}$$

the last inequality again comes from fact that n is sufficiently large. If the claim holds, then not all candidates in $W_{k^*, \log n}$ can be covered by $\text{Im}(\psi_{t^*-1})$, contradicting the assumption that the embedding strategy fails at time t^* .

It therefore remains to prove the claim. We proceed by induction on t .

For the base case $t = 1$, the embedding strategy embeds v_1 into some vertex of $W_{k,1}$, since $L^-(v_1) = \emptyset$. Hence $|\text{Im}(\psi_1) \cap W_{k,1}| = 1 \leq L_{k,1}$, and $|\text{Im}(\psi_1) \cap W_{k,j}| = 0 < L_{k,j}$ for every $j > 1$. Thus the claim holds for $t = 1$.

Now let $1 < t < t^* - 1$, and assume that $|\text{Im}(\psi_{t-1}) \cap W_{k,j}| \leq L_{k,j}$ for every k and j . We will show that $|\text{Im}(\psi_t) \cap W_{k,j}| \leq L_{k,j}$ for every k and j .

Suppose not. Since the only difference between $\text{Im}(\psi_t)$ and $\text{Im}(\psi_{t-1})$ is the newly embedded vertex $\psi_t(v_t)$. Therefore there exists a unique pair (k', j') such that $\psi_t(v_t) \in W_{k', j'}$ and

$$|\text{Im}(\psi_t) \cap W_{k', j'}| > L_{k', j'}. \quad (\text{A1})$$

For every other pair $(k, j) \neq (k', j')$, we still have

$$|\text{Im}(\psi_t) \cap W_{k,j}| \leq L_{k,j}. \quad (\text{A2})$$

We shall show that this is impossible.

Case 1: $j' = 1$.

If $k' = N$, then

$$|\text{Im}(\psi_t) \cap W_{N,1}| > L_{N,1} = n,$$

which is impossible since H has only n vertices and $t \leq n$.

Now suppose $k' < N$. Recall that a vertex of H is embedded into $W_{k'}$ only if its degree is greater than $\Delta_{k'}$. By Observation 2.1 in [Kat25], the number of such vertices is at most $\frac{rn}{\Delta_{k'}}$. Therefore

$$|\text{Im}(\psi_t) \cap W_{k',1}| \leq \frac{rn}{\Delta_{k'}} = L_{k',1},$$

contradicting to assumption A1.

Case 2: $j' > 1$.

Define

$$\mathcal{B} = \left\{ B_x : B_x = \psi_t(L^-(x)), \psi_t(x) \in W_{k', j'} \right\}.$$

Notice that $|\mathcal{B}| = |\text{Im}(\psi_t) \cap W_{k', j'}|$.

We claim that $\mathcal{B}$ is well-behaved. Condition (WB1) holds since every back-link has size at most 1, and ψ_t is an embedding.

For (WB2), let $u \in W_k$. Suppose $u \in B_i$ for some i . Then there exists a vertex $v_b \in V(H)$ such that $u \in \psi_t(L^-(v_b))$. Since $u \in \text{Im}(\psi_t)$, there exists a vertex v_a with $\psi_t(v_a) = u$. As u belongs to the embedded back-link of v_b , we have $a < b$. Therefore

$$|\{i : u \in V(B_i)\}| \leq d_H(v_a) \leq \Delta_{k-1}$$

by the definition of embedding strategy. Hence (WB2) holds.

For (WB3), by the induction hypothesis we have $|\text{Im}(\psi_{t-1}) \cap W_{k,j}| \leq L_{k,j} + 1$ for every k, j . We shall show that $L_{k,j} + 1 \leq \frac{1}{16}|W_{k,j}|$ for every k, j .

First consider $j = 1$. If $k = N$, then $|W_{N,1}| = \frac{1}{2}|W_N| \geq 50rn$. Since $L_{N,1} = n$, $L_{N,1} + 1 = n + 1 < \frac{50}{16}rn \leq \frac{1}{16}|W_{N,1}|$ for sufficiently large n . Now suppose $k < N$. Then $|W_{k,1}| = \frac{1}{2}|W_k| = \frac{50 \cdot 3^q rn}{\Delta_k}$, and $L_{k,1} + 1 = \frac{rn}{\Delta_k} + 1 < \frac{50 \cdot 3^q rn}{16\Delta_k} = \frac{1}{16}|W_{k,1}|$ for sufficiently large n .

Now consider $j > 1$. Since $L_{k,j} = (4 \log n)^{1-j} L_{k,1} = \frac{L_{k,1}}{4 \log n}$, and $|W_{k,j}| \geq \frac{|W_{k,1}|}{\log n}$, we obtain

$$L_{k,j} + 1 \leq \frac{L_{k,1}}{4 \log n} + 1 \leq \frac{|W_{k,1}|}{64 \log n} + 1 \leq \frac{|W_{k,j}|}{64} + 1$$

Furthermore, $|W_{k,j}| \geq \frac{50 \cdot 3^q rn}{\Delta_k \log n} = \frac{50 \cdot 3^q r n^{1-q-k}}{\log n}$, which tends to infinity as $n \rightarrow \infty$. Hence, for sufficiently large n , $\frac{|W_{k,j}|}{64} + 1 \leq \frac{1}{16}|W_{k,j}|$. Therefore $L_{k,j} + 1 \leq \frac{1}{16}|W_{k,j}|$, and (WB3) follows.

Thus $\mathcal{B}$ is well-behaved. We may therefore apply Lemma 2 to $\mathcal{B}$ with $k = k'$ and $j = j' - 1$.

If $L_{k',j'-1} \geq 1$, then Lemma 2 gives

$$\begin{aligned} \left| \{u \in W_{k',j'-1} : \exists B \in \mathcal{B}, B \subseteq L(u)\} \right| &\geq \min \left\{ \frac{1}{16}, \frac{1}{8} |\mathcal{B}| p^* \Delta_{k'} \right\} |W_{k',j'-1}| \\ &\geq \min \left\{ \frac{|W_{k',j'-1}|}{16}, \frac{1}{8} |\text{Im}(\psi_t) \cap W_{k',j'}| p^* \Delta_{k'} |W_{k',j'-1}| \right\} \\ &\geq \min \left\{ L_{k',j'-1} + 1, \frac{1}{8} L_{k',j'} p^* \Delta_{k'} |W_{k',j'-1}| \right\} \\ &\geq \min \left\{ L_{k',j'-1} + 1, \frac{1}{8} \frac{L_{k',j'-1}}{4 \log n} \cdot \frac{K_r (\log n)^2}{n} \cdot \Delta_{k'} \cdot \frac{100 \cdot 3^q rn}{2 \Delta_{k'} \log n} \right\} \\ &\geq \min \left\{ L_{k',j'-1} + 1, \frac{100 \cdot 3^q r K_r}{64} L_{k',j'-1} \right\} \end{aligned} \quad (3)$$

As $L_{k',j'-1} \geq 1$, either term in equation 3 is at least $L_{k',j'-1} + 1$. However assumption A2 assumes the number of candidates cover by $\mathfrak{S}\phi_t$ is at most $L_{k',j'-1}$, which means there is at least one candidate available.

If $L_{k',j'-1} < 1$, we have $|\mathcal{B}| = |\text{Im}(\psi_t) \cap W_{k',j'}| \geq 1$, as $\psi_t(v_t) \in W_{k',j'}$.

Applying Lemma 2 to $\mathcal{B}$ with $k = k'$ and $j = j' - 1$, we obtain

$$\left| \{u \in W_{k',j'-1} : \exists B \in \mathcal{B}, B \subseteq L(u)\} \right| \geq \min \left\{ \frac{1}{16}, \frac{1}{8} |\mathcal{B}| p^* \Delta_{k'} \right\} |W_{k',j'-1}| \geq \min \left\{ \frac{1}{16}, \frac{1}{8} p^* \Delta_{k'} \right\} |W_{k',j'-1}|$$

We claim that both terms inside the minimum are at least 1:

$$\frac{1}{16} |W_{k',j'-1}| \geq \frac{1}{16} \cdot \frac{50 \cdot 3^q rn}{\Delta_{k'} \log n} = \frac{50 \cdot 3^q r}{16} \cdot \frac{n^{1-q-k'}}{\log n} > 1$$

for sufficiently large n .

Moreover,

$$\frac{1}{8} p^* \Delta_{k'} |W_{k',j'-1}| \geq \frac{1}{8} \cdot \frac{K_r (\log n)^2}{n} \cdot \Delta_{k'} \cdot \frac{50 \cdot 3^q rn}{\Delta_{k'} \log n} = \frac{50 \cdot 3^q r K_r}{8} \log n > 1$$

for sufficiently large n .

Therefore $\left| \{u \in W_{k',j'-1} : \exists B_i \in \mathcal{B}, B_i \subseteq L(u)\} \right| \geq 1$, which means that there exists a candidate for some x in $W_{k',j'-1}$ whose back-link is B_i . Since $L_{k',j'-1} < 1$, the induction hypothesis gives $|\text{Im}(\psi_{t-1}) \cap W_{k',j'-1}| = 0$. Thus this candidate is available. This contradicts the definition of j' as the smallest integer such that $W_{k',j'}$ contains an available candidate for x .

This contradiction completes the proof of the claim, and hence the proof of the theorem. $\square$

References

[Kat25] Peter Allen Julia Böttcher Jasmin Katz. Bounds for hypergraph universality. 2025.